

Task 15: Window Functions for Advanced Analytics

Tools:

- Primary: MySQL 8+ / PostgreSQL
- Alternatives: Google BigQuery Sandbox, SQLite (limited support)

Hints / Mini Guide:

1. Extend the employees table by adding salary, joining date, and department information.
2. Use ROW_NUMBER() to assign unique ranks within each department.
3. Apply RANK() and DENSE_RANK() to understand ranking differences.
4. Calculate running totals using SUM() OVER(PARTITION BY ...).
5. Use LAG() and LEAD() to compare salaries between consecutive employees.
6. Apply window functions without collapsing rows unlike GROUP BY.
7. Combine WHERE clauses with window functions using subqueries.
8. Map analytics use cases like salary trends and performance analysis.

Deliverables:

- SQL file with multiple window function queries

Final Outcome:

- Intern can write advanced analytical SQL queries used in data platforms.

Interview Questions Related To Above Task:

- What is a window function?
- Difference between RANK and DENSE_RANK?
- Why window functions don't reduce rows?
- What is PARTITION BY?
- Real-world use cases of LAG and LEAD?

📌 Task Submission Guidelines

■ 🕒 Time Window:

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

■ 🔍 Self-Research Allowed:

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

■ 🛠️ Debug Yourself:

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

■ 💰 No Paid Tools:

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

■ 📁 GitHub Submission:

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

📌 Submit Here:

After completing the task, paste your GitHub repo link and submit it using the link below:

- 🖱️ [[Submission Link](#)]

Best
of
Luck

